

Setup Sheet for Program 1001

JOB DESCRIPTION: Setup2
DOCUMENT PATH: C:\Users\HP\Downloads\Part3.ipt

Setup

WCS: #0

STOCK:

DX: 23mm

DY: 23mm

DZ: 161mm

PART:

DX: 19mm

DY: 19mm

DZ: 130mm

STOCK LOWER IN WCS #0:

X: -11.5mm

Y: -11.5mm

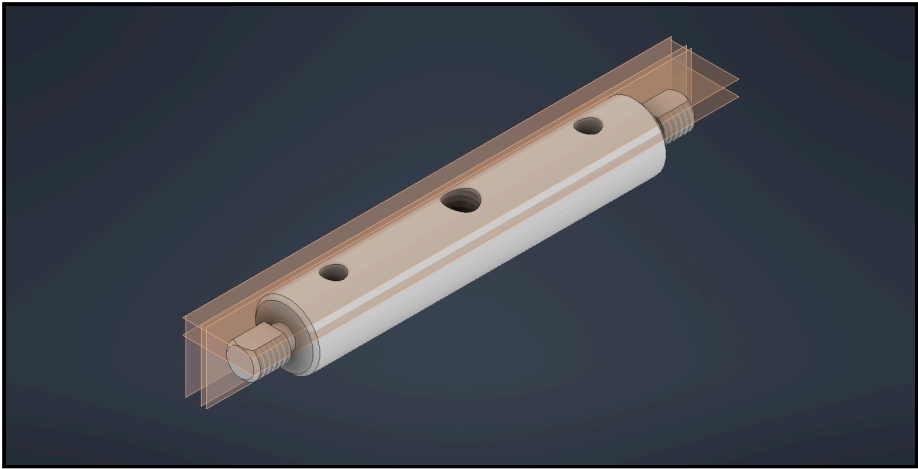
Z: -161mm

STOCK UPPER IN WCS #0:

X: 11.5mm

Y: 11.5mm

Z: 0mm



Total

NUMBER OF OPERATIONS: 18

NUMBER OF TOOLS: 9

TOOLS: T1 T2 T3 T8 T9 T10 T11 T12 T13

MAXIMUM FEEDRATE: 1600mm/min

MAXIMUM SPINDLE SPEED: 5000rpm

CUTTING DISTANCE: 1279.51mm

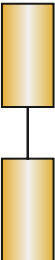
RAPID DISTANCE: 2523.37mm

ESTIMATED CYCLE TIME: 18m:49s

Tools			
T1 D1			
TYPE: groove turning	TOOL ORIENTATION IN TURRET: Radial	HOLDER: External Right	
INSERT: Square	MAXIMUM FEED: 56.841mm/min		
WIDTH: 3mm	MAXIMUM SPINDLE SPEED: 1600rpm		
COMPENSATION: Tip tangent	CUTTING DISTANCE: 34.5mm		
	RAPID DISTANCE: 228.5mm		
	ESTIMATED CYCLE TIME: 2m:9s (11.4%)		
T2 D2 L2			
TYPE: center drill LIVE	MAXIMUM FEED: 160mm/min		
DIAMETER: 10mm	MAXIMUM SPINDLE SPEED: 1600rpm		
TIP ANGLE: 118°	CUTTING DISTANCE: 26mm		
LENGTH: 50mm	RAPID DISTANCE: 155mm		
FLUTES: 3	ESTIMATED CYCLE TIME: 41s (3.6%)		

file:///C:/Users/HP/Desktop/Part3.html

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T3 D0 TYPE: general turning INSERT: ISO C 80deg EDGE LENGTH: 4mm NOSE RADIUS: 0.8mm CROSS SECTION: T TOLERANCE: M RELIEF: A3deg COMPENSATION: Tip tangent DESCRIPTION: CNMT Right Hand	TOOL ORIENTATION IN TURRET: Radial MAXIMUM FEED: 176.903mm/min MAXIMUM SPINDLE SPEED: 1600rpm CUTTING DISTANCE: 618.23mm RAPID DISTANCE: 1109.38mm ESTIMATED CYCLE TIME: 9m:39s (51.3%)	HOLDER: ISO L Right 
T8 D8 L8 TYPE: right hand tap LIVE DIAMETER: 10mm LENGTH: 50mm FLUTES: 3	MAXIMUM FEED: 375mm/min MAXIMUM SPINDLE SPEED: 250rpm CUTTING DISTANCE: 48mm RAPID DISTANCE: 24mm ESTIMATED CYCLE TIME: 8s (0.7%)	
T9 D9 L9 TYPE: drill LIVE DIAMETER: 8.5mm TIP ANGLE: 118° LENGTH: 110.5mm FLUTES: 3	MAXIMUM FEED: 135mm/min MAXIMUM SPINDLE SPEED: 900rpm CUTTING DISTANCE: 35.18mm RAPID DISTANCE: 56.93mm ESTIMATED CYCLE TIME: 16s (1.4%)	
T10 D10 L10 TYPE: drill LIVE DIAMETER: 6mm TIP ANGLE: 118° LENGTH: 78mm FLUTES: 3	MAXIMUM FEED: 127.5mm/min MAXIMUM SPINDLE SPEED: 1275rpm CUTTING DISTANCE: 68.86mm RAPID DISTANCE: 167.36mm ESTIMATED CYCLE TIME: 34s (3%)	
T11 D11 L11 TYPE: flat end mill LIVE DIAMETER: 16mm LENGTH: 80mm FLUTES: 3	MAXIMUM FEED: 126mm/min MAXIMUM SPINDLE SPEED: 700rpm CUTTING DISTANCE: 59.95mm RAPID DISTANCE: 60.3mm ESTIMATED CYCLE TIME: 40s (3.6%)	
T12 D0 TYPE: thread turning INSERT: ISO double COMPENSATION: Tip tangent DESCRIPTION: OD Threading	TOOL ORIENTATION IN TURRET: Radial MAXIMUM FEED: 0mm/min MAXIMUM SPINDLE SPEED: 5000rpm CUTTING DISTANCE: 231.43mm RAPID DISTANCE: 453.28mm ESTIMATED CYCLE TIME: 5s (0.5%)	HOLDER: Straight Right 
T13 D13 TYPE: groove turning INSERT: Square WIDTH: 3mm NOSE RADIUS: 0.2mm COMPENSATION: Tip tangent	TOOL ORIENTATION IN TURRET: Radial MAXIMUM FEED: 1600mm/min MAXIMUM SPINDLE SPEED: 1600rpm CUTTING DISTANCE: 157.36mm RAPID DISTANCE: 268.63mm ESTIMATED CYCLE TIME: 51s (4.5%)	HOLDER: External Right 

Operations

Operation 1/18

DESCRIPTION: Face2
STRATEGY: Turning Face
WCS: #0
TOLERANCE: 0.01mm
MAXIMUM STEPOVER: 0.5mm

SURFACE SPEED: 30m/min
FEEDRATE PER REV: 0.1mm
CUTTING DISTANCE: 33.69mm
RAPID DISTANCE: 41.86mm
ESTIMATED CYCLE TIME: 36s (3.2%)
COOLANT: Flood

T3 D0

TYPE: general turning
INSERT: ISO C 80deg
EDGE LENGTH: 4mm
NOSE RADIUS: 0.8mm
CROSS SECTION: T
TOLERANCE: M
RELIEF: A3deg
COMPENSATION: Tip tangent
DESCRIPTION: CNMT Right Hand



Operation 2/18

DESCRIPTION: Drill4
STRATEGY: Drilling
WCS: #0
TOLERANCE: 0.01mm

MAXIMUM SPINDLE SPEED: 1600rpm
MAXIMUM FEEDRATE: 160mm/min
CUTTING DISTANCE: 8mm
RAPID DISTANCE: 30mm
ESTIMATED CYCLE TIME: 3s (0.3%)
COOLANT: Flood

T2 D2 L2

TYPE: center drill LIVE
DIAMETER: 10mm
TIP ANGLE: 118°
LENGTH: 50mm
FLUTES: 3



Operation 3/18

DESCRIPTION: Profile Roughing3
WCS: #0
TOLERANCE: 0.01mm
MAXIMUM STEPOVER: 1.2mm

SURFACE SPEED: 30m/min
FEEDRATE PER REV: 0.15mm
CUTTING DISTANCE: 336.18mm
RAPID DISTANCE: 369.28mm
ESTIMATED CYCLE TIME: 4m:32s (24.1%)
COOLANT: Flood

T3 D0

TYPE: general turning
INSERT: ISO C 80deg
EDGE LENGTH: 4mm
NOSE RADIUS: 0.8mm
CROSS SECTION: T
TOLERANCE: M
RELIEF: A3deg
COMPENSATION: Tip tangent
DESCRIPTION: CNMT Right Hand



Operation 4/18

DESCRIPTION: Profile Finishing2
WCS: #0
TOLERANCE: 0.01mm
MAXIMUM STEPOVER: 1.2mm

SURFACE SPEED: 35m/min
FEEDRATE PER REV: 0.07mm
CUTTING DISTANCE: 126.27mm
RAPID DISTANCE: 155.21mm
ESTIMATED CYCLE TIME: 2m:54s (15.4%)
COOLANT: Flood

T3 D0

TYPE: general turning
INSERT: ISO C 80deg
EDGE LENGTH: 4mm
NOSE RADIUS: 0.8mm
CROSS SECTION: T
TOLERANCE: M
RELIEF: A3deg
COMPENSATION: Tip tangent
DESCRIPTION: CNMT Right Hand



Operation 5/18

DESCRIPTION: Single Groove1
STRATEGY: Turning Groove
WCS: #0
TOLERANCE: 0.01mm
STOCK TO LEAVE: 0mm

SURFACE SPEED: 25m/min
FEEDRATE PER REV: 0.05mm
CUTTING DISTANCE: 18mm
RAPID DISTANCE: 56mm
ESTIMATED CYCLE TIME: 1m:9s (6.1%)
COOLANT: Flood

T1 D1

TYPE: groove turning
INSERT: Square
WIDTH: 3mm
COMPENSATION: Tip tangent



Operation 6/18

DESCRIPTION: Face3
STRATEGY: Facing
WCS: #0
TOLERANCE: 0.01mm
MAXIMUM STEPOVER: 15.2mm

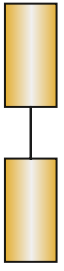
MAXIMUM SPINDLE SPEED: 700rpm
MAXIMUM FEEDRATE: 126mm/min
CUTTING DISTANCE: 30.73mm
RAPID DISTANCE: 30.9mm
ESTIMATED CYCLE TIME: 21s (1.8%)
COOLANT: Flood

T11 D11 L11

TYPE: flat end mill LIVE
DIAMETER: 16mm
LENGTH: 80mm
FLUTES: 3



Operation 7/18 DESCRIPTION: Thread2 STRATEGY: Turning Thread WCS: #0 TOLERANCE: 0.01mm	MAXIMUM SPINDLE SPEED: 450rpm FEEDRATE PER REV: 1mm CUTTING DISTANCE: 133.72mm RAPID DISTANCE: 183.14mm ESTIMATED CYCLE TIME: 2s (0.2%) COOLANT: Flood	T12 D0 TYPE: thread turning INSERT: ISO double COMPENSATION: Tip tangent DESCRIPTION: OD Threading	
Operation 8/18 DESCRIPTION: Drill4 (2) STRATEGY: Drilling WCS: #0 TOLERANCE: 0.01mm	MAXIMUM SPINDLE SPEED: 500rpm MAXIMUM FEEDRATE: 30mm/min CUTTING DISTANCE: 18mm RAPID DISTANCE: 125mm ESTIMATED CYCLE TIME: 38s (3.3%) COOLANT: Flood	T2 D2 L2 TYPE: center drill LIVE DIAMETER: 10mm TIP ANGLE: 118° LENGTH: 50mm FLUTES: 3	
Operation 9/18 DESCRIPTION: Drill6 STRATEGY: Drilling WCS: #0 TOLERANCE: 0.01mm	MAXIMUM SPINDLE SPEED: 900rpm MAXIMUM FEEDRATE: 135mm/min CUTTING DISTANCE: 35.18mm RAPID DISTANCE: 56.93mm ESTIMATED CYCLE TIME: 16s (1.4%) COOLANT: Flood	T9 D9 L9 TYPE: drill LIVE DIAMETER: 8.5mm TIP ANGLE: 118° LENGTH: 110.5mm FLUTES: 3	
Operation 10/18 DESCRIPTION: Drill7 STRATEGY: Drilling WCS: #0 TOLERANCE: 0.01mm	MAXIMUM SPINDLE SPEED: 1275rpm MAXIMUM FEEDRATE: 127.5mm/min CUTTING DISTANCE: 68.86mm RAPID DISTANCE: 167.36mm ESTIMATED CYCLE TIME: 34s (3%) COOLANT: Flood	T10 D10 L10 TYPE: drill LIVE DIAMETER: 6mm TIP ANGLE: 118° LENGTH: 78mm FLUTES: 3	
Operation 11/18 DESCRIPTION: Drill8 STRATEGY: Drilling WCS: #0 TOLERANCE: 0.01mm	MAXIMUM SPINDLE SPEED: 250rpm MAXIMUM FEEDRATE: 375mm/min CUTTING DISTANCE: 48mm RAPID DISTANCE: 24mm ESTIMATED CYCLE TIME: 8s (0.7%) COOLANT: Flood	T8 D8 L8 TYPE: right hand tap LIVE DIAMETER: 10mm LENGTH: 50mm FLUTES: 3	
Operation 12/18 DESCRIPTION: Groove5 STRATEGY: Turning Profile Groove WCS: #0 TOLERANCE: 0.01mm STOCK TO LEAVE: 0mm/1mm	SURFACE SPEED: 96m/min FEEDRATE PER REV: 0.08mm CUTTING DISTANCE: 157.36mm RAPID DISTANCE: 268.63mm ESTIMATED CYCLE TIME: 51s (4.5%) COOLANT: Flood	T13 D13 TYPE: groove turning INSERT: Square WIDTH: 3mm NOSE RADIUS: 0.2mm COMPENSATION: Tip tangent	
Operation 13/18 DESCRIPTION: Face2 (2) STRATEGY: Turning Face WCS: #0 TOLERANCE: 0.01mm MAXIMUM STEPOVER: 0.5mm	SURFACE SPEED: 30m/min FEEDRATE PER REV: 0.1mm CUTTING DISTANCE: 30.69mm RAPID DISTANCE: 156.86mm ESTIMATED CYCLE TIME: 31s (2.7%) COOLANT: Flood	T3 D0 TYPE: general turning INSERT: ISO C 80deg EDGE LENGTH: 4mm NOSE RADIUS: 0.8mm CROSS SECTION: T TOLERANCE: M RELIEF: A3deg COMPENSATION: Tip tangent DESCRIPTION: CNMT Right Hand	

Operation 14/18 DESCRIPTION: Profile Roughing3 (2) WCS: #0 TOLERANCE: 0.01mm MAXIMUM STEPOVER: 1.2mm	SURFACE SPEED: 30m/min FEEDRATE PER REV: 0.15mm CUTTING DISTANCE: 68.47mm RAPID DISTANCE: 219.28mm ESTIMATED CYCLE TIME: 43s (3.8%) COOLANT: Flood	T3 D0 TYPE: general turning INSERT: ISO C 80deg EDGE LENGTH: 4mm NOSE RADIUS: 0.8mm CROSS SECTION: T TOLERANCE: M RELIEF: A3deg COMPENSATION: Tip tangent DESCRIPTION: CNMT Right Hand	
Operation 15/18 DESCRIPTION: Profile Finishing2 (2) WCS: #0 TOLERANCE: 0.01mm MAXIMUM STEPOVER: 1.2mm	SURFACE SPEED: 35m/min FEEDRATE PER REV: 0.07mm CUTTING DISTANCE: 22.94mm RAPID DISTANCE: 166.88mm ESTIMATED CYCLE TIME: 23s (2%) COOLANT: Flood	T3 D0 TYPE: general turning INSERT: ISO C 80deg EDGE LENGTH: 4mm NOSE RADIUS: 0.8mm CROSS SECTION: T TOLERANCE: M RELIEF: A3deg COMPENSATION: Tip tangent DESCRIPTION: CNMT Right Hand	
Operation 16/18 DESCRIPTION: Single Groove1 (2) STRATEGY: Turning Groove WCS: #0 TOLERANCE: 0.01mm STOCK TO LEAVE: 0mm	SURFACE SPEED: 25m/min FEEDRATE PER REV: 0.05mm CUTTING DISTANCE: 16.5mm RAPID DISTANCE: 172.5mm ESTIMATED CYCLE TIME: 1m:1s (5.4%) COOLANT: Flood	T1 D1 TYPE: groove turning INSERT: Square WIDTH: 3mm COMPENSATION: Tip tangent	
Operation 17/18 DESCRIPTION: Face3 (2) STRATEGY: Facing WCS: #0 TOLERANCE: 0.01mm MAXIMUM STEPOVER: 15.2mm	MAXIMUM SPINDLE SPEED: 700rpm MAXIMUM FEEDRATE: 126mm/min CUTTING DISTANCE: 29.23mm RAPID DISTANCE: 29.4mm ESTIMATED CYCLE TIME: 20s (1.7%) COOLANT: Flood	T11 D11 L11 TYPE: flat end mill LIVE DIAMETER: 16mm LENGTH: 80mm FLUTES: 3	
Operation 18/18 DESCRIPTION: Thread2 (2) STRATEGY: Turning Thread WCS: #0 TOLERANCE: 0.01mm	MAXIMUM SPINDLE SPEED: 450rpm FEEDRATE PER REV: 1mm CUTTING DISTANCE: 97.72mm RAPID DISTANCE: 270.14mm ESTIMATED CYCLE TIME: 3s (0.3%) COOLANT: Flood	T12 D0 TYPE: thread turning INSERT: ISO double COMPENSATION: Tip tangent DESCRIPTION: OD Threading	